

Kamal Swarnkar

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SKILLS

Languages:	Python, C, JavaScript
Frontend:	React, HTML, CSS
Backend:	Django, Django REST Framework
Databases:	PostgreSQL, Redis
Tools:	Git, Linux, Docker, Postman, VSCode, Celery, JWT, SolidWorks, OpenRocket
Libraries & ML:	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, spaCy, NLTK
Deployment:	Vercel, Render, Railway
Soft Skills:	Self-motivated, Articulate, Organized, Leadership

PROJECTS

AI Career Lab | GitHub | Live Demo

Mar' 2026

- Reduced analysis latency by **~97% (5.8s → ~180ms)** by offloading LLM-based enrichment to **Celery + Redis asynchronous workers**, improving responsiveness significantly.
- Scaled system throughput by **7–12× (30 → 500 req/min)** by decoupling ML inference from external API calls and serving predictions independently of enrichment.
- Achieved **86.6% classification accuracy (F1: 0.86)** with minimal overfitting (1.23% gap) using **TF-IDF** features and optimized Scikit-learn classification models.
- Improved model reliability and generalization by implementing a structured ML pipeline with evaluation across **RF, SVM, and Logistic Regression** and real-time inference.
- Reduced deployment overhead and improved response efficiency by building a **lightweight Django-based backend** with server-rendered templates and optimized asset delivery.
- **Tech Stack:** Python, Django, Scikit-learn, Pandas, NumPy.

Resumix - Resume Analysis & Role Prediction Platform | GitHub | Live Demo

Feb' 2026

- Reduced API response time by **~90% (3.2s → ~400ms)** by migrating resume parsing and AI enrichment to **Celery + Redis asynchronous** processing pipelines.
- Improved query performance by **35–45%** through PostgreSQL **indexing** strategies and **Django ORM prefetching**, reducing latency in history and analytics endpoints.
- Optimized ML inference latency by **~85% (16ms → ~2ms)** via **in-memory model loading** instead of repeated per-request deserialization using joblib.
- Designed a **scalable backend using Django REST Framework** and implemented JWT-based stateless authentication for secure and efficient API access.
- Engineered a **multi-factor scoring pipeline** combining **TF-IDF, skills, ATS**, and experience to enable **automated gap detection** and personalized recommendations.
- **Tech Stack:** Python, Django, PostgreSQL, JWT, Scikit-learn, XGBoost, spaCy, NLTK, Celery, Redis.

TRAINING

FLAMES '25 MERN with Gen AI | W3grads | Certificate

Jul' 2025

- Resolved development bottlenecks in **AI workflows** and database performance by optimizing pipelines and debugging inconsistencies, improving **system reliability** and deployment stability.
- Improved team efficiency by establishing a structured **Git workflow** and enforcing **daily SCRUMs**, reducing merge conflicts and ensuring consistent collaboration across distributed contributors.
- Enhanced application scalability by **addressing deployment and integration challenges** in **full-stack systems**, enabling smoother releases and faster iteration cycles.

Youth Research Program | Resolute Lab | Certificate

Mar' 2025

- Designed and simulated rocket systems using **OpenRocket** and **SolidWorks**, performing propulsion analysis and **payload integration studies** to validate engineering feasibility.
- Optimized design parameters through **iterative simulations** and applied **aerospace principles** to **improve structural stability** and performance of high-power rocket models.
- Collaborated in a **multidisciplinary team** to develop and document a **complete rocket system**, ensuring accurate reporting and alignment with research and simulation objectives.

CERTIFICATES

- Master Generative AI & Generative AI tools (Infosys) | Certificate Aug' 2025
- Build Generative AIApps and Solutions with No-Code Tools | Certificate Aug' 2025
- The Bits and Bytes of Computer Networking (Google) | Certificate Sep' 2024
- Computer Communications (University of Colorado) | Certificate Sep' 2024

ACHIEVEMENTS

- Qualified GATE 2026 (CS/IT) in first attempt with a score of 383 with AIR 21411.
- Solved 100+ DSA Problems covering arrays, hashing, trees, recursion, and optimization techniques.
- MEDHA 2024 Hackathon: Achieved 10th rank among 60 teams.

EDUCATION

Lovely Professional University	Phagwara, Punjab
• Bachelor of Technology - Computer Science and Engineering – CGPA: 8.90	Aug' 2023 – Present
Central Academy Senior Secondary School	Bhilwara, Rajasthan
• Intermediate – 91%	Apr' 2021 – Mar'2022
• Matriculation – 89%	Apr' 2019 – Mar'2020